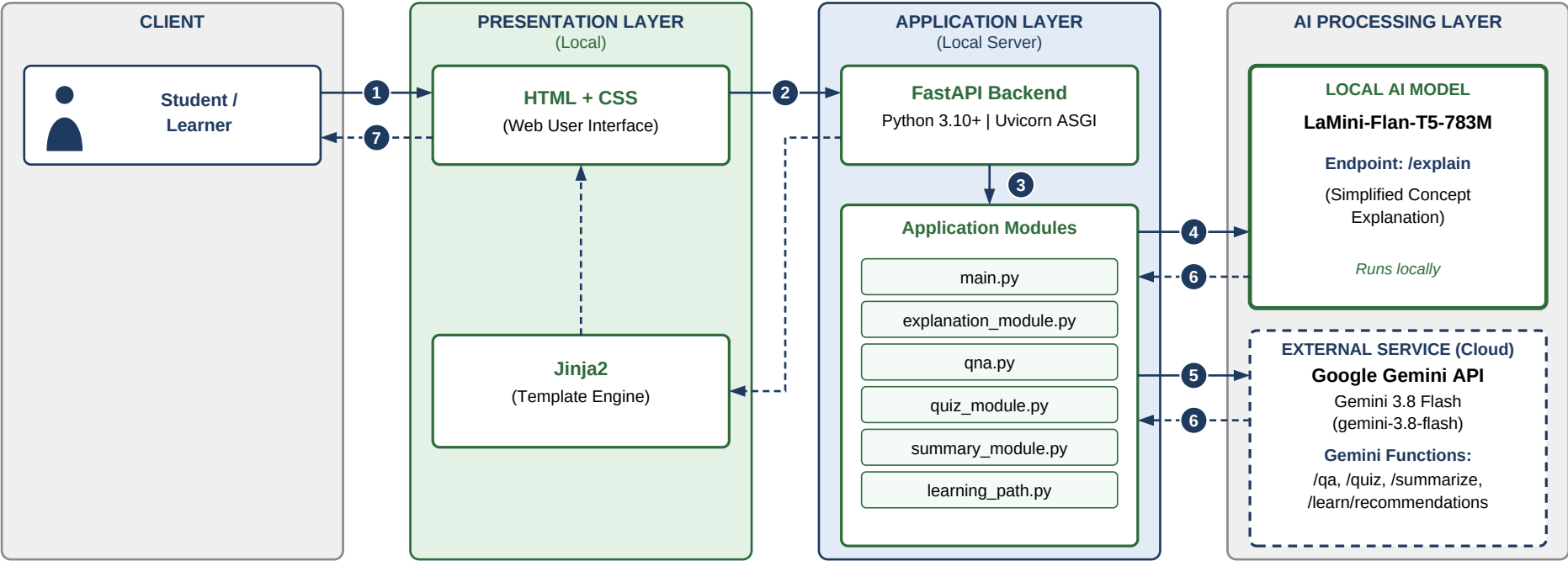


Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	1 October 2026
TEAM ID	6ab7f80ad083b0e2ee72b17e
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Architecture Diagram



→ Request flow ←----- Response flow (returned via FastAPI, Jinja2, HTML)

1 Student submits request 2 UI forwards to FastAPI 3 FastAPI routes to the feature module 4 /explain handled by local LaMini-Flan-T5-783M
5 /qa, /quiz, /summarize, /learn/recommendations sent to Gemini API 6 AI output returned to FastAPI 7 Rendered via Jinja2 + HTML and shown to student

No database / cloud storage used in the prototype

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web interface through which the student interacts with EduGenie	HTML, CSS
2	Template Engine	Renders dynamic content and AI responses into HTML pages	Jinja2
3	Application Logic-1 (Request Handling)	Handles HTTP requests, routing and feature modules	FastAPI, Python 3.10+
4	Application Logic-2 (ASGI Server)	Serves the FastAPI application locally	Uvicorn
5	Explanation	Simplified concept explanation (/explain), generated locally	LaMini-Flan-T5-783M
6	Q&A	Answers student questions (/qa)	Gemini API, Gemini 3.8 Flash
7	Quiz	Generates quizzes (/quiz)	Gemini API, Gemini 3.8 Flash
8	Summarization	Summarizes study content (/summarize)	Gemini API, Gemini 3.8 Flash
9	Learning Recommendations	Suggests learning paths and next topics (/learn/recommendations)	Gemini API, Gemini 3.8 Flash
10	Database	No database used in the prototype	Not Used / N/A
11	Cloud Database	No cloud database used in the prototype	Not Used / N/A
12	File Storage	Application files stored on the local machine	Local Filesystem
13	External API	Cloud AI service for Q&A, quiz, summary and recommendations	Google Gemini API
14	ML Model-1	Local open-source model for concept explanation	LaMini-Flan-T5-783M
15	ML Model-2	Cloud-hosted model accessed through the Gemini API	Google Gemini 3.8 Flash
16	Infrastructure	Application runs on a local machine, with no cloud hosting	Local Server / Development Environment

Table 2: Application Characteristics

S.No	Characteristic	Implemented in the Prototype	Recommended (Not Currently Implemented)
1	Open-Source Frameworks	FastAPI, Uvicorn and Jinja2 (Python); open-source LaMini-Flan-T5-783M model for local explanations.	Pin dependency versions and scan them regularly for known vulnerabilities.
2	Security Implementations	Gemini API calls are made from the FastAPI backend rather than the browser; no user data is stored (no database).	Keep the API key in environment variables or a secret manager; enable HTTPS; add input validation, rate limiting and user authentication.
3	Scalable Architecture	Modular design (one module per feature) on an asynchronous ASGI stack (FastAPI + Uvicorn).	Containerize with Docker, run multiple Uvicorn workers behind a load balancer, and deploy to a cloud platform.
4	Availability	Single local server instance; /explain works without an external service, while Gemini features need internet access.	Cloud hosting with health checks and redundancy; retry and fallback handling for Gemini API failures.
5	Performance	Local model avoids a network call for /explain; Gemini Flash model chosen for fast responses; asynchronous request handling.	Cache frequent responses, use GPU or quantized inference for the local model, and monitor latency.

Table 3: AI Model Responsibility

AI Model	Deployment	Endpoint(s)	Responsibility
LaMini-Flan-T5-783M	Local	/explain	Simplified concept explanation
Gemini 3.8 Flash (gemini-3.8-flash)	External (Google Gemini API)	/qa, /quiz, /summarize, /learn/recommendations	Question answering, quiz generation, summarization and learning recommendations

Technology Stack Summary

Layer / Component	Technology
Frontend	HTML/CSS
Template Engine	Jinja2
Backend	FastAPI
Language	Python 3.10+
ASGI Server	Uvicorn
Local AI Model	LaMini-Flan-T5-783M
Cloud AI	Google Gemini API
Gemini Model	gemini-3.8-flash